A PROJECT REPORT

ON

WINE QUALITY PREDICTION USING DECISION TREES AND RANDOM FOREST

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

Computer Science and Information Technology (CSIT)

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by

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"Wine Quality Prediction Using Decision Trees and Random Forest."

1. Abstract:

Wine quality assessment is an important task in the food and beverage industry because it helps manufacturers maintain consistent product standards and improve customer satisfaction. Traditionally, wine quality is evaluated by human experts through sensory analysis, which can be time-consuming, expensive, and subjective. With the advancement of Machine Learning, it has become possible to predict wine quality automatically using the chemical properties of wine.

This project focuses on predicting wine quality using two popular Machine Learning algorithms: Decision Tree and Random Forest. The dataset used for this project contains various physicochemical attributes of wine, such as fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates, and alcohol content. These attributes are used as input features to predict the quality score of the wine.

The project begins with data collection and preprocessing, where missing values, duplicate records, and inconsistent data are handled. Exploratory Data Analysis (EDA) is performed to understand the relationships between different features and wine quality. The dataset is then divided into training and testing sets. The Decision Tree algorithm creates a tree-like model to classify wine quality, while the Random Forest algorithm builds multiple decision trees and combines their predictions to improve accuracy and reduce overfitting.

The performance of both models is evaluated using metrics such as accuracy, precision, recall, F1-score, and confusion matrix. The results are compared to determine which algorithm performs better for wine quality prediction. The Random Forest model is expected to achieve higher accuracy due to its ensemble learning approach.

This project demonstrates how Machine Learning techniques can automate quality prediction, reduce human effort, and assist the wine industry in maintaining high production standards. The developed model can serve as a reliable decision-support tool for predicting wine quality based on measurable chemical characteristics.

2. Introduction:

Wine is one of the most widely consumed beverages around the world, and maintaining its quality is essential for both producers and consumers. The quality of wine depends on several physicochemical characteristics such as acidity, sugar content, alcohol percentage, pH value, sulphates, and density. Traditionally, wine quality is evaluated by experienced tasters through sensory analysis. Although effective, this process is subjective, time-consuming, and may produce inconsistent results due to differences in human judgment.

Machine Learning provides an efficient alternative by analyzing historical data and identifying patterns that influence wine quality. By training predictive models on existing wine datasets, it is possible to estimate the quality of new wine samples accurately without relying solely on human experts.

This project aims to develop a wine quality prediction system using Decision Tree and Random Forest algorithms. Decision Tree is a simple and interpretable algorithm that predicts outcomes by splitting data into branches based on feature values. Random Forest is an ensemble learning technique that combines multiple Decision Trees to improve prediction accuracy and reduce overfitting.

The project involves collecting the wine quality dataset, preprocessing the data, performing exploratory data analysis, training the Machine Learning models, evaluating their performance, and comparing the results. Various evaluation metrics such as accuracy, precision, recall, F1-score, and confusion matrix are used to assess model performance.

The proposed system provides a fast, accurate, and cost-effective method for predicting wine quality. It helps reduce manual effort, supports decision-making in the wine industry, and demonstrates the practical application of Machine Learning in quality assessment. The knowledge gained from this project can also be extended to similar quality prediction problems in manufacturing, food processing, and agriculture.

3. Software Requirements:

Operating System	Windows 10/11, Linux, or macOS
Programming Language	Python 3.9 or above
ML Libraries	Jupyter Notebook, VS Code, or PyCharm
Data Visualization	Scikit-learn
Web Framework (Backend)	Pandas
Frontend Technologies	NumPy
Database	Matplotlib, Seaborn
IDE / Code Editor	Wine Quality Dataser (CSV)
Version Control	Git
Model Deployment	Flask or Streamlit (for deployment
Browser	Google Chrome, Microsoft Edge, Mozilla Firefox, or any modern web browse

Signature of the Guide **Course Instructor**